Lec 11

Saturday, 31 May 2025 09.20

1. Calculate the inverse z-transform of the following sequences:
   a) $X(z) = (1+3z)(1+2z^{-1})(1-z^{-1})$
   b) $X(z) = \frac{1-\frac{1}{2}z^{-1}}{1+\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2}}$ $|z| > \frac{1}{2}$
   c) $X(z) = \frac{1-\frac{1}{2}z^{-1}}{1-\frac{1}{4}z^{-2}}$ $|z| > \frac{1}{2}$
   d) $X(z) = \frac{1-az^{-1}}{z^{-1}-a}$ $|z| > \left|\frac{1}{a}\right|$
   e) $X(z) = \ln(1-4z)$ $|z| < \frac{1}{4}$
   f) $X(z) = \frac{3-7z^{-1}+5z^{-2}}{1-\frac{5}{2}z^{-1}+z^{-2}}$ $\frac{1}{2} < |z| < 2$

a) $X(z) = \underbrace{(1+3z)\cdot(1+2z^{-1})}\cdot(1-z^{-1})$ (ligner meget finite-length sequence eksemplet fra bogen (S. 117 example 3.10, second edition))

$= (1+3z+2z^{-1}+6)\cdot(1-z^{-1})$

$= 1+3z+2z^{-1}+6-z^{-1}-3-2z^{-2}-6z^{-1}$

$= 4+3z-5z^{-1}-2z^{-2}$

$\Downarrow z^{-1}$ (tabel opslag (4.))

$x[n] = 4\delta[n]+3\delta[n+1]-5\delta[n-1]-2\delta[n-2]$

b)

$X(z) = \frac{1-\frac{1}{2}z^{-1}}{1+\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2}}$ $|z| > \frac{1}{2}$

Partial fraction: Vil gerne have det på denne form

$X(z) = \sum_{k=1}^{N} \frac{A_k}{1-d_k z^{-1}}$

Navneren er ikke på den generale form af:

$(1+az^{-1})(1+bz^{-1})$

Derfor ganges parenteserne ud.

$(1+az^{-1})(1+bz^{-1}) = 1+(a+b)z^{-1}+(ab)z^{-2}$

For så at finde ud af hvad a og b så er, så sætter vi dem bare op som 2 ligninger og 2 ukendte.

$a+b = \frac{3}{4}$

$a\cdot b = \frac{1}{8}$

Få maple til at gøre det, hvis man ikke gider at spilde tiden

$f := a+b = \frac{3}{4}$

$g := a\cdot b = \frac{1}{8}$

Det har ikke nogen betydning vilken en af dem man assigner, resultatet vil bare omvendt men stadig gyldig

$\text{Solve}(\{f,g\},\{a,b\}) = \{a=\frac{1}{2}, b=\frac{1}{4}\},\{a=\frac{1}{4}, b=\frac{1}{2}\}$

med dette har vi så vores Poly. til at være:

$X(z) = \frac{1-\frac{1}{2}z^{-1}}{(1+\frac{1}{4}z^{-1})\cdot(1+\frac{1}{2}z^{-1})} = \frac{A_1}{1+\frac{1}{4}\cdot z^{-1}} + \frac{A_2}{1+\frac{1}{2}\cdot z^{-1}}$ $|z| > \frac{1}{2}$

Nu skal vi lige finde z:

- $1+\frac{1}{4}z^{-1} = 0 \Rightarrow z = -\frac{1}{4}$
- $1+\frac{1}{2}z^{-1} = 0 \Rightarrow z = -\frac{1}{2}$

Nu kan partial fraction bruges

$$X(z) = \sum_{k=1}^{N} \frac{A_k}{1-d_k z^{-1}} \quad , d_k = z$$

A_k findes ved:

$$A_k = (1-d_k z^{-1}) \cdot X(z) \Big|_{z=d_k}$$

$$A_1 = \cancel{(1+\tfrac{1}{4}z^{-1})} \cdot \frac{1-\frac{1}{2}z^{-1}}{\cancel{(1+\frac{1}{4}z^{-1})} \cdot (1+\frac{1}{2}z^{-1})} \Bigg|_{z=-\frac{1}{4}} = -3$$

$$A_2 = (1+\tfrac{1}{2}z^{-1}) \cdot \frac{1-\frac{1}{2}z^{-1}}{(1+\frac{1}{4}z^{-1}) \cdot (1+\frac{1}{2}z^{-1})} \Bigg|_{z=-\frac{1}{2}} = 4$$

$$X(z) = \frac{-3}{1+\frac{1}{4}\cdot z^{-1}} + \frac{4}{1+\frac{1}{2}\cdot z^{-1}}$$

Husk at dette er d_k

$|z| > \frac{1}{2}$ (Begge er en right-sided sequence) (Bruger (5))

$$X(z) \xrightarrow{z^{-1}} x[n] = -3\left(-\tfrac{1}{4}\right)^n u[n] + 4\left(\tfrac{1}{2}\right)^n u[n]$$

c)

c) $X(z) = \dfrac{1-\frac{1}{2}z^{-1}}{1-\frac{1}{4}z^{-2}} \qquad |z| > \frac{1}{2}$

$$X(z) = \frac{\cancel{1-\frac{1}{2}z^{-1}}}{(1+\frac{1}{2}z^{-1})\cancel{(1-\frac{1}{2}z^{-1})}} = \frac{1}{1+\frac{1}{2}z^{-1}}$$

$1+\frac{1}{2}z^{-1} \Rightarrow z = -\frac{1}{2}$

Since $|z| > \frac{1}{2}$, the sequence is right sided →

Tabel opslag (5)

$$x[n] = \left(-\tfrac{1}{2}\right)^n u[n]$$

Han har splittet det op i dens positive og negative composanter.

Vi vil gerne factorisere:

$1-\frac{1}{4}z^{-2}$

det vil vi så gerne skrive som: $a^2-b^2 = (a+b)(a-b)$

1. $a = 1$, $b = \frac{1}{4}z^{-2}$ $\quad (\frac{1}{2})^2 = \frac{1}{4}$

2. Apply identity:

$$1-\tfrac{1}{4}z^{-2} = 1^2 - \left(\tfrac{1}{2}z^{-1}\right)^2$$

$$a^2-b^2 = (a+b)(a-b)$$

$$1^2 - \left(\tfrac{1}{2}z^{-1}\right)^2 = \left(1+\tfrac{1}{2}z^{-1}\right)\left(1-\tfrac{1}{2}z^{-1}\right)$$

Maple:

$(1+bz^{-1})(1-bz^{-1}) = 1-\frac{1}{4}z^{-2} \Rightarrow b = \{\frac{1}{2}, -\frac{1}{2}\}$

d) $X(z) = \dfrac{1-az^{-1}}{z^{-1}-a} \qquad |z| > \left|\frac{1}{a}\right|$

Kan splittes

$$\frac{1}{a-b} = \frac{1}{b}\cdot\frac{1}{1-\frac{a}{b}}$$

$$\frac{a}{x-a} = \frac{1}{\frac{x}{a}-1}$$

Mellenregning, forklaring $\quad \dfrac{az^{-1}}{z^{-1}-a} = \dfrac{z^{-1}}{\frac{z^{-1}}{a}-1} = \dfrac{1}{\frac{1}{a}z^{-1}-1}\, z^{-1}$

$$X(z) = \frac{1}{z^{-1}-a} - \frac{az^{-1}}{z^{-1}-a} = -\frac{1}{a}\,\frac{1}{1-\frac{1}{a}z^{-1}} - \frac{z^{-1}}{\frac{1}{a}z^{-1}-1} =$$

$$= -\frac{1}{a}\,\frac{1}{1-\frac{1}{a}z^{-1}} + \frac{1}{1-\frac{1}{a}z^{-1}}\cdot z^{-1}$$

$$\frac{1}{a}\,\frac{1}{1-\frac{1}{a}z^{-1}} - \frac{1}{\frac{1}{a}z^{-1}-1} = \frac{1}{a}\,\frac{1}{1-\frac{1}{a}z^{-1}} - \left(\frac{1}{-1+\frac{1}{a}z^{-1}}\right)$$

$$= \frac{1}{a}\,\frac{1}{1-\frac{1}{a}z^{-1}} + \frac{1}{1-\frac{1}{a}z^{-1}}$$

Since $|z| > \left|\frac{1}{a}\right|$, both sequences are right-sided →

$$X[n] = -\frac{1}{a}\left(\frac{1}{a}\right)^n u[n] + \left(\frac{1}{a}\right)^{n-1} u[n-1] =$$

$$= -\left(\frac{1}{a}\right)^{n+1} u[n] + \left(\frac{1}{a}\right)^{n-1} u[n-1]$$

e) $X(z) = \ln(1-4z) \qquad |z| < \frac{1}{4}$

We know that

$$\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$$

} Reglen kan ses i bogen S. 117, Second edition eq.(3.51) $X(z) = \log(1+az^{-1}) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} a^n z^{-n}}{n}$

We then have

$$\ln(1-4z) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(-4z)^n}{n} = \sum_{n=1}^{\infty} (-1) \frac{(-1)^n (-4)^n z^n}{n} =$$

$$= -\sum_{n=1}^{\infty} \frac{4^n}{n} z^n = -4z - \frac{4^2}{2} z^2 - \frac{4^3}{3} z^3 - \ldots$$

Therefore:

$$x[n] = -4\delta[n+1] - \frac{4^2}{2}\delta[n+2] - \frac{4^3}{3}\delta[n+3] + \ldots =$$ } Tabel opslag (4)

$$= -\frac{4^n}{n} u[-n-1]$$ } Tabel opslag (6) Tror dette er en generalisering af (4)

f)

$$X(z) = \frac{3 - 7z^{-1} + 5z^{-2}}{1 - \frac{5}{2}z^{-1} + z^{-2}} \qquad \frac{1}{2} < |z| < 2$$

Dette kan ses som:

$$X(z) = \frac{M}{N} = \frac{3 - 7z^{-1} + 5z^{-2}}{1 - \frac{5}{2}z^{-1} + z^{-2}}$$

da M = N (Samme størrelses polynomiel) så bruges:

$$X(z) = B_0 \frac{A_1}{1 - d_k z^{-1}} + \frac{A_2}{1 - d_k z^{-1}}$$

B_0 kan findes ved long division:

$N \overline{)\,M}$ Der står reelt 1 der. 1 går op i 5, 5 gange

$$\begin{array}{r|l} & 5 \\ \underbrace{z^{-2} - \frac{5}{2}z^{-1} + 1}_{N} & 5z^{-2} - 7z^{-1} + 3 \\ N\cdot 5 \rightarrow & 5z^{-2} - \frac{25}{2}z^{-1} + 5 \\ \hline & \frac{11}{2}z^{-1} - 2 \end{array}$$

* Husk at polynomiet skal aranges i rækkefølge af z størelserne (flip poly)

* Dette bliver vores nye tæller a.k.a (M)

$$X(z) = \frac{\frac{11}{2}z^{-1} - 2}{\underbrace{1 - \frac{5}{2}z^{-1} + z^{-2}}}$$

Bliver nu splittet op, ligesom i opg. b

Vil gerne have på formen: $(1+az^{-1})(1-az^{-1})$

$$\left.\begin{array}{l} a+b = -\frac{5}{2} \\ a \cdot b = 1 \end{array}\right\} \left\{a = -2, b = -\frac{1}{2}\right\}$$

$(1 - 2z^{-1})(1 - \frac{1}{2}z^{-1})$

$$X_1(z) = \frac{\frac{11}{2}z^{-1} - 2}{(1 - 2z^{-1})(1 - \frac{1}{2}z^{-1})}$$

$1 - 2z^{-1} = 0 \Rightarrow z = d_k = 2$

$1 - \frac{1}{2}z^{-1} = 0 \Rightarrow z = d_k = \frac{1}{2}$

$$X(z) = B_0 \frac{A_1}{1 - 2z^{-1}} + \frac{A_2}{1 - \frac{1}{2}z^{-1}}$$

$$A_1 = (1 - d_k \cdot z^{-1}) \cdot X_1(z) \Big|_{z=2}$$

$$= \cancel{(1 - 2 \cdot z^{-1})} \cdot \frac{\frac{11}{2} z^{-1} - 2}{\cancel{(1 - 2 z^{-1})} (1 - \frac{1}{2} z^{-1})} \Bigg|_{z=2} = 1$$

$$A_2 = (1 - d_k \cdot z^{-1}) \cdot X_1(z) \Big|_{z=\frac{1}{2}}$$

$$= \cancel{(1 - \frac{1}{2} \cdot z^{-1})} \cdot \frac{\frac{11}{2} z^{-1} - 2}{(1 - 2 z^{-1}) \cancel{(1 - \frac{1}{2} z^{-1})}} \Bigg|_{z=\frac{1}{2}} = -3$$

$$X(z) = 5 + \frac{1}{1 - 2\ z^{-1}} + \frac{-3}{1 - \frac{1}{2}\ z^{-1}}$$

z^{-1} (1.) (6.) (5.)

$$x[n]\ 5\delta[n] - 2^n u[-n-1] - 3 \cdot \left(\frac{1}{2}\right) u[n]$$